As a library, NLM provides access to scientific literature. Inclusion in an NLM database does not imply endorsement of, or agreement with, the contents by NLM or the National Institutes of Health.
Learn more: PMC Disclaimer | PMC Copyright Notice

World Journal of Clinical Cases

World J Clin Cases. 2021 Jul 6;9(19):4881–4889. doi: 10.12998/wjcc.v9.i19.4881

# Fear of missing out: A brief overview of origin, theoretical underpinnings and relationship with mental health

Mayank Gupta [1], Aditya Sharma [2]

Author information Article notes Copyright and License information

PMCID: PMC8283615 PMID: 34307542

## Abstract

Fear of missing out (FoMO) is a unique term introduced in 2004 to describe a phenomenon observed on social networking sites. FoMO includes two processes; firstly, perception of missing out, followed up with a compulsive behavior to maintain these social connections. We are interested in understanding the complex construct of FoMO and its relations to the need to belong and form stable interpersonal relationships. It is associated with a range of negative life experiences and feelings, due to it being considered a problematic attachment to social media. We have provided a general review of the literature and have summarized the findings in relation to mental health, social functioning, sleep, academic performance and productivity, neuro-developmental disorders, and physical well-being. We have also discussed the treatment options available for FoMo based on cognitive behavior therapy. It imperative that new findings on FoMO are communicated to the clinical community as it has diagnostic implications and could be a confounding variable in those who do not respond to treatment as usual.

**Keywords:** Fear of missing out, Mental health, Physical well-being, Academic performance, Fear of missing out-reduction, Problematic social media

**Core Tip:** Fear of missing out is a relatively new construct that needs the attention of the clinical community due to its diagnostic implications in treatment processes. It is associated to problematic social media usage and can be experienced as a range of negative emotions and feelings related to the need to belong.

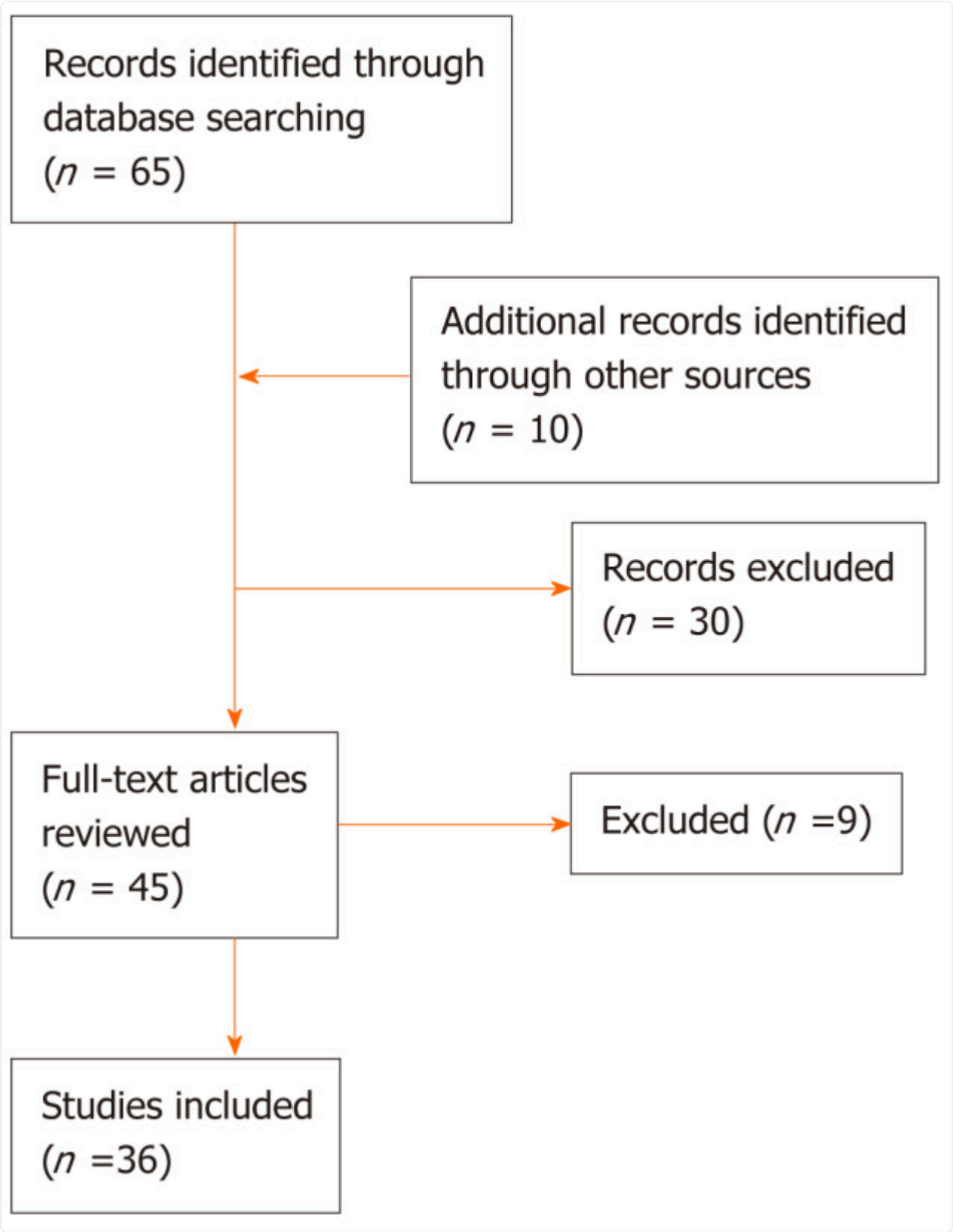
## INTRODUCTION

Fear of missing out (FoMO) is a unique term introduced in 2004 and then extensively used since 2010[1] to describe a phenomenon observed on social networking sites. It eventually made it to the oxford dictionary in 2013[2]. In 2013[3] British psychologists elaborated and defined it as "pervasive apprehension that others might be having rewarding experiences from which one is absent", FoMO is characterized by the desire to stay continually connected with what others are doing. FoMO was conceptualized using self-determination theory (SDT), which was developed by Ryan *et al*[4] and applied by Przybylski *et al*[3] to understanding what drives FOMO. In SDT[5], social relatedness can drive intrinsic motivation, which in turn can encourage positive mental health. Przybylski *et al*[3] applied SDT to FoMO, proposing that FoMO is a negative emotional state resulting from unmet social relatedness needs. The conceptualization that FoMO involves negative affect from unmet social needs is similar to theories about the negative emotional effects of social ostracism[6]. FoMO is a relatively new psychological phenomenon. It may exist as an episodic feeling that occurs in mid-conversation, as a long-term disposition, or a state of mind that leads the individual to feel a deeper sense of social inferiority, loneliness, or intense rage[7].Today, more than ever, people are exposed to a lot of details about what others are doing; and people are faced with the continuous uncertainty about whether they are doing enough or if they are where they should be in terms of their life[8]. FoMO includes two processes; firstly, perception of missing out, followed up with a compulsive behavior to maintain these social connections. The social aspect of FoMO could be postulated as relatedness which refers to the need to belong, and formation of strong and stable interpersonal relationships[9]. FoMO is considered as a type of problematic attachment to social media, and is associated with a range of negative life experiences and feelings, such as a lack of sleep, reduced life competency, emotional tension, negative effects on physical well-being, anxiety and a lack of emotional control; with intimate connections possibly being seen as a way to counter social rejection[10]. In the last seven years research is done to establish its links with various mental health outcomes. We are interested in understanding which population is more vulnerable to this phenomenon. It is also unclear which reward pathway seems to be involved in its reinforcing effects. It has been widely accepted that dopaminergic tracts, in particular, mesolimbic systems get activated with successful social connections[11]. The reward prediction error encoding and variable reward schedules are known to maintain these behaviors. FoMO is not a unitary phenomenon, but rather a more complex construct that could reflect a certain personal predisposition and maintained by rewarding experiences that results from people's desire for interpersonal attachments.

## REVIEW OF THE LITERATURE

This paper provides a general review of the literature. A literature search was conducted in PubMed, Google Scholar databases with coverage from January 2002 to December 2020. The search terms included "FoMO," "depression," "anxiety," "sleep," "academic" and "physical well-being." Results were restricted to peer-reviewed English-language articles. Conference and dissertation proceedings, case reports, protocol papers and opinion pieces were excluded. Reference lists were examined for additional articles of relevance. We identified $n = 65$ articles according to the above search methods. An additional 10 articles were identified through hand search and Google Scholar articles. Total $n = 30$ was excluded according to abstract screening. Full text review was performed for $n = 45$ articles according to study inclusion criteria. The $n = 9$ articles failed to meet stated inclusion criteria. Figure 1 shows the flowchart for study selection for inclusion.

Figure 1.




Open in a new tab

The flowchart for study selection for inclusion.

Social networking sites (SNS) provides a compensatory medium for adolescents with social anxiety[12] to address their unmet social needs in a manner other than face to face communication[13]. The use of SNS contributes to easier communication[14] for those with deficits by compensating for their unmet social needs with much less effort and instantaneously[15,16]. However, this "social compensation" can be problematic when it reinforces avoidance for face-to-face and consequentially increasing social anxiety[17]. These processes likely to worsen social fears and predisposes one for anxiety disorders. FoMO is also associated with problematic SNS use due to its easy access for adolescents to interact at will[18,19] and constant need for personal validation and rewarding appraisals of distorted sense of self[20]. FoMO cognitive aspect is manifested by negative ruminations like frequently checking and refreshing SNS for alerts and notifications. These subsequently heightens the levels of anxiety in order to keep up with the theme with anticipation of a reward[21]. The concept of FoMO explores the fear of social exclusion. Through social media, there is continuous awareness of what an individual may be missing in terms of a good time which researcher phrases as "it creates distorted perceptions of edited lives of others". The 'Round the clock' nature of these communication may lead to feeling lonely and inadequate through highlighting others activities and popularity and comparison of oneself to others, leading to vicious cycle of compulsive checking and engagement. FoMO has a relationship with the amount time spent on SNS as a predictor of emotional distress[22]. The constant "upward social comparisons" and unreasonable expectations can adversely impact one's self-esteem. These events are associated with emergence of depressive symptoms in some individuals[23]. These depressive symptoms may be further compounded by perceptions that one can avoid negative emotions when part of these online communities[24]. In a Belgian[25] study with 1000 subjects, 6.5% were found to be using SNSs excessively, they were found to have lower emotional stability and agreeableness, conscientiousness, perceived control and self-esteem which could be risk factor for affective disorders. The problematic use of internet and development of FoMO is associated with young adults ignoring peer relationships, potentially leading to depressive symptoms[26]. Additionally, the longer time spent using SNS (spending more than 2 h per day) demonstrated a significantly higher risk of having suicidality[27]. FOMO may have a mediational role between narcissism and problematic SNS use suggesting that unmet social relatedness needs cause high engagement in problematic SNS use[28,29]. A constant need for rewarding experiences reinforces one with FoMO to engage in risky activities[30] to maximize socialization opportunities[3]. FoMO has been associated with negative alcohol-related consequences either through higher alcohol use or greater willingness to engage in higher risk behaviors[31]. Adolescents with FoMO may likely to experiment with drugs and alcohol[32] to fit in with peers on SNS.

## FOMO AND SOCIAL FUNCTIONING

Despite the instantaneous and desired interactions with peers *via* social media, young adults are feeling lonelier and more disconnected than ever[33]. FoMO may exacerbate pre-existing feelings of loneliness after engaging on SNSs extensively[34]. It is argued that communication channels with fewer nonverbal cues may result in less warmth and closeness among those who are interacting in verbal means, avoiding meaningful and pragmatic communication[35]. This may result in misinterpretations and misunderstanding leading to further emotional dissatisfaction and feelings of loneliness[36]. It is believed that time spent on social networking sites, activates the amygdala and the fear pathway, rendering young adults vulnerable to feelings of loneliness, and social disconnectedness[3]. This contributes to a cyclical nature[3] of FoMO where an individual engages with SNS to alleviate feelings of loneliness, but in fact exacerbates them. They have to again return to SNS and attempt to alleviate these feelings[37]. Conversely, young adults who experienced less satisfaction of the basic

psychological needs for competence, autonomy, and connectedness to others, also reported higher levels of FoMO[3]. FoMO broadcasts more options than can be pursued, impacting an individual's decision making in personal and professional settings[38]. It may impair an individual's ability to make commitments and agreements, as one feels inclined to keep options open and not risk losing out on an important, potentially life-changing experience which may offer greater meaning and personal gratification.

## FOMO AND RELATIONSHIP WITH SLEEP

The FoMO and interpersonal stress are associated with insomnia and subsequently poor mental health outcomes[39]. In another Chinese study of university students, researchers found negative affect, which is "a general dimension of subjective distress and unpleasurable engagement that subsumes a variety of aversive mood states, including anger, contempt, disgust, guilt, fear, and nervousness", was linked with poor sleep mediated by FoMO[40]. In an Israeli university study of 40 participants who measured smartphone use at night, were at risk of reduced sleep quality and overall psychological health[41]. A survey of 101 adolescents linked pre-sleep worry and FoMO to longer sleep onset latency and reduced sleep duration[42]. It is also well established in clinical research, the blue light emitting from the screen of the electronic devices (short-wave blue light with wavelength between 415 nm and 455 nm) affect sleep[43]. The postulated mechanism includes suppression of pineal hormone melatonin leading to a state of neuropsychological arousal. These alterations in circadian rhythm may lead to increased sleep latency, reduced sleep duration, and increased daytime sleep.

## ACADEMIC PERFORMANCE AND PRODUCTIVITY

FoMO is closely related to SNS use and has negative effects on academic performance[44].The effects of problematic internet usage (PIU) and responding to frequent notifications entails repeated switching between tasks. This affects attention span; interrupts work and overall productivity[45]. The constant connection *via* smartphones can impair cognitive abilities and cause academic distractions. Furthermore, rapid switching between tasks is related to poor learning[46]. It is established multitaskers make more mistakes and take longer to complete tasks[47]. Additionally, overuse of SNS has been associated with decreased academic performance and activity[48].

## FOMO AND NEURODEVELOPMENTAL DISORDERS

There were no current studies which link patients who have attention deficit hyperactivity disorder (ADHD) with FoMO. However, it will be of interest to study children and adolescents with hyperactivity and inattention who already have delayed myelination in prefrontal cortex[49]. It could be a reasonable hypothesis based on the recent evidence, that these patients would be more vulnerable to FoMO. And subsequently have further worsening in areas of self-evaluation, self-monitoring, and the interpretation of affect in social situations[50,51]. Further research is needed in this area to explore variables likely to be affected with emergence of FoMO.

## FOMO AND PHYSICAL WELLBEING

The young adults with high levels of FoMO are less likely to report their own lifestyle as healthy. The feelings of envy and social exclusion are also linked with poor eating habits[52]. Additionally, FoMO promotes high SNS use which leads to a sedentary lifestyle influencing the epidemic of obesity in young adults[53]. The amount of time spent on SNS is associated with vision issues[54] and poor attention, which is linked with accidents while walking, crossing roads, driving or doing daily activities. PIU due to FoMO may lead to poor posture and subsequent musculoskeletal pain involving neck, back and hands and fingers[55]. This is further accentuated as pain prevents the initiation or the continuation of sleep, contributing to poor physical health.

## TREATMENT OPTIONS FOR FOMO

The treatment choice is based on the principles of cognitive behavior therapy (CBT)[56]. CBT addresses distorted cognitions which is postulated to be underlying its development and maintenance of these conditions[57]. It also addresses underlying anxiety and focuses on the predisposing factors[56]. The main treatment goal should be control rather than abstinence. FoMO reduction (FoMO-R)[57] is a novel promising specifically designed model aimed at building resilience by focusing on self-help literacy guide about design and appropriate use of SNS[58]. It is an active, effective user-friendly, safe alternative to anxiolytic drugs and self-help technique relying on dealing with the vicious circle model of anxiety[59]. The method associates between the context of use, fears and technical countermeasures. The effectiveness of cognitive reappraisal of anxiety has been recognized in helping people recognize and manage their digital addiction. Compulsive behaviors can be regulated using techniques such as distraction and reappraisal. Distraction avoidance techniques are also a part of FoMO-R. Excessive involvement in the targeted behaviors leading to self-reported functional impairment tends to be fairly transient for most individuals hence the excessive behaviors such as FoMO are often context-dependent, and that spontaneous recovery is frequent[60]. Individuals are also offered alternative strategies and opportunities for connecting with other people without having the feeling of missing out on something. Some studies have reported use of anxiolytics and found it helpful.

## DISCUSSION

As per CDC, child and teens ages 8-18 spend an average of 7.5 h[61] on screen including Television. PIU[62] has been on the rise and its association with child and adolescent psychopathology has been well established. Since the introduction of SNS in 2008, there has been a steady rise in increased social interactions online. In 2014, the term FoMO which was initially used in marketing was formally adapted in clinical psychiatry to describe a unique phenomenon. It's evident based on recent research that it's a complex psychological underpinning involving cognitive, behavioral and addiction processes. FoMO may start with distorted thinking related to sense of fear of being left out from a rewarding experience. However, it is reinforced with constant responsiveness to SNS. Some research also terms these behaviors as compulsive. We continue to doubt if these behaviors are of obsessive nature, but clearly constant rechecking and browsing of SNS is hallmark. These behaviors are impairing several domains as discussed in the earlier part of this paper. We also hypothesize that the complex nature FoMO could be due to rapid change in technology in last two decades. Social[63] anthropology studies have demonstrated how SNS may be impacting social norms. However, it would be a fair argument that human cognitive processes have been lagging behind a rapidly changing technology interface. These researches are an attempt to recognize and understand origins and psychological interplay with multiple

brain process. FoMO has been linked with not only distractibility, but overall decline in productivity and worse mental health outcomes. The recent studies have established association with sleep disturbances, social anxiety, clinical depression and decline in academic performance. It's not clear who all are vulnerable to FoMO, but its observed that certainly personality traits and one with underlying mental health problems could be more affected. The measurement scale validated for FoMO has established in empirical research that these sequelae could be extremely impairing and debilitating. We did not observe any research linking FoMO with ADHD. ADHD is a disorder characterized with deficits with delayed gratification and theoretically one suffering from ADHD would not only be vulnerable but also likely to have worse overall outcomes related to FoMO. ADHD is strongly associated with internet gaming disorder[64] and we postulate that these subtypes would be more prone to aggressive outbursts. An alternative explanation of these phenomenon could be attributed to the negative feelings of envy[65]. The most accepted reasons are lack of intimacy in these interactions and having a large group on social media where one constantly finds themselves to compare oneself with others[66]. In this pool, it is likely people will experience frustration and envy from comparing oneself to others but still tempted in striving to be closer to the person of comparison[67]. These behaviors could exacerbate negative emotions such as envy, jealousy, resentment, and anxiety as well as the desire to chase social media perfection[68]. It is likely the effects will be worsened for individuals with mental health issues[69]. The knowledge of these underlying psychological processes would be crucial for providing psychoeducation and also planning individualized psychotherapy-based interventions. It's also interesting to note that FoMO is a global phenomenon, and research evidence is present from North America, Asia, Europe and Australia which further underscores the need to understand the address. With regards to treatment, the common theme of all the research was its origin lies in the amount of screen time. After the recommended screen time is exceeded, one is more vulnerable to access information and maintain social relationships *via* SNS or social gaming networks. The key preventive measure remains restricting to the recommended screen time. The minimum age for using SNS is set at 13 and more legislative measures are needed to raise the age to 16. However, it has been reported that many younger children also have accounts on SNS. Parental education also has been found to be a highly effective measure to curb this. The American Academy of Pediatrics recommended Family Media Use Plan[70] has been a useful tool. Beside prevention, promising research based on transtheoretical model has been an effective. It's interesting to note that a key feature of this intervention is making one understand how social media works.

## CONCLUSION

FoMO is a now an established entity in the research community. However, many practicing clinicians are not aware or educated about it affecting their population. It's imperative these new findings are communicated to the clinical community. Given it has both diagnostic implications and also could be a confounding variable in ones who do not respond to the treatment as usual. There is need for further research which includes prevalence studies, psychological understanding and more evidence-based prevention and treatment interventions.

## Footnotes

Conflict-of-interest statement: All authors declare that they have no competing interests.

Manuscript source: Invited manuscript

Peer-review started: February 6, 2021

First decision: February 28, 2021

Article in press: May 19, 2021

Specialty type: Psychiatry

Country/Territory of origin: United States

Peer-review report's scientific quality classification

Grade A (Excellent): 0

Grade B (Very good): 0

Grade C (Good): C

Grade D (Fair): 0

Grade E (Poor): 0

P-Reviewer: Seeman MV S-Editor: Wang JL L-Editor: A P-Editor: Zhang YL

## Contributor Information

Mayank Gupta, Clarion Psychiatric Center and Department of Child and Adolescent Psychiatry, Lake Erie College of Osteopathic Medicine, Clarion, PA 16214, United States.

Aditya Sharma, Department of Neuroscience, University of Pittsburgh, Pittsburgh, PA 15213, United States. aas142@pitt.edu.

## References

1. Morford M, Gate Columnist SF. Oh My God You Are so Missing Out. Available from: https://www.sfgate.com/entertainment/morford/article/Oh-my-God-you-are-so-missing-out-2536241.php .

2. Oxford English Online Dictionary. FOMO. Available from: http://www.oxforddictionaries.com/us/definition/american_english/FOMO .

3. Przybylski AK, Murayama K, DeHaan CR, Gladwell V. Motivational, emotional, and behavioral correlates of fear of missing out. Comput Human Behav. 2013;29:1841–1848. [Google Scholar]

4. Ryan RM, Deci EL. Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. Am Psychol. 2000;55:68–78. doi: 10.1037//0003-066x.55.1.68. [DOI] [PubMed] [Google Scholar]

5. Koole SL, Schlinkert C, Maldei T, Baumann N. Becoming who you are: An integrative review of self-determination theory and personality systems interactions theory. J Pers. 2019;87:15–36. doi: 10.1111/jopy.12380. [DOI] [PMC free article] [PubMed] [Google Scholar]

6. Williams KD. Ostracism. Annu Rev Psychol. 2007;58:425–452. doi: 10.1146/annurev.psych.58.110405.085641. [DOI] [PubMed] [Google Scholar]

7. Zaslove M. What is the psychology behind FOMO? Available from: https://www.quora.com/What-is-the-psychology-behind-FOMO .

8. Rifkin J, Cindy C, Barbara Kahn B. Fomo: How the Fear of Missing Out Leads to Missing Out, in NA - Advances in Consumer Research Volume 43. Duluth: Association for Consumer Research, 2015: 244-248. [Google Scholar]

9. La Guardia JG, Patrick H. Self-determinationtheory as a fundamental theory of close relationships. Can Psychol . 2008;49:201. [Google Scholar]

10. Altuwairiqi M, Jiang N, Ali R. Problematic Attachment to Social Media: Five Behavioural Archetypes. Int J Environ Res Public Health. 2019;16 doi: 10.3390/ijerph16122136. [DOI] [PMC free article] [PubMed] [Google Scholar]

11. The Guardian, Guardian News and Media. Has Dopamine Got Us Hooked on Tech? Available from: http://www.theguardian.com/technology/2018/mar/04/has-dopamine-got-us-hooked-on-tech-facebook-apps-addiction .

12. Clayton RB, Osborne RE, Miller BK, Oberle CD. Loneliness, anxiousness, and substance use as predictors of Facebook use. Comput Human Behav. 2013;29:687–693. [Google Scholar]

13. Bhagat S. Is Facebook a planet of lonley individuals? Int J Indian Psychology. 2015;3:5–9. [Google Scholar]

14. Baker LR, Oswald DL. Shyness and online social networking services. J Soc Pers Relat . 2010;27:873–889. [Google Scholar]

15. Huang LY, Hsieh YJ, WuYCJ Gratifications and social network service usage: The mediating role of online experience. Info Manag 2014; 51: 774-782, DOI: 10.1016/j.im.2014.05. 4:00. [Google Scholar]

16. Song H, Zmyslinski-Seelig A, Kim J, Drent A, Victor A, Omori K, Allen M. Does Facebook make you lonely? Comput Human Behav. 2014;36:446–452. [Google Scholar]

17. Kuss DJ, Griffiths MD. Online social networking and addiction--a review of the psychological literature. Int J Environ Res Public Health. 2011;8:3528–3552. doi: 10.3390/ijerph8093528. [DOI] [PMC free article] [PubMed] [Google Scholar]

18. Stead H, Bibby PA, Personality , fear of missing out and problematic internet use and their relationship to subjective well-being. Comput Human Behav. 2017;76:534–540. [Google Scholar]

19. Strickland A. Exploring the effects of social media use on the mental health of young adults. Available from: https://stars.library.ucf.edu/honorstheses1990-2015/1684/

20. Burrow AL, Rainone N. How many likes did I get? J Exp Soc Psychol. 2017;69:232–236. [Google Scholar]

21. Billieux J, Philippot P, Schmid C, Maurage P, De Mol J, Van der Linden M. Is Dysfunctional Use of the Mobile Phone a Behavioural Addiction? Clin Psychol Psychother. 2015;22:460–468. doi: 10.1002/cpp.1910. [DOI] [PubMed] [Google Scholar]

22. Weinstein A, Dorani D, Elhadif R, Bukovza Y, Yarmulnik A, Dannon P. Internet addiction is associated with social anxiety in young adults. Ann Clin Psychiatry. 2015;27:4–9. [PubMed] [Google Scholar]

23. Steers MN, Wickham RE, Acitelli LK. Seeing everyone else's highlight reels: How Facebook usage is linked to depressive symptoms. J Soc Clin Psychol. 2014;33:701–731. [Google Scholar]

24. Twenge J, Joiner T, Rogers M, Martin G. Increases in depressive symptoms, suicide related outcomes, and suicide rates among U.S. adolescents after 2010 and links to increased new media screen time. Clin Psychol Sci. 2017;6:3–17. [Google Scholar]

25. De Cock R, Vangeel J, Klein A, Minotte P, Rosas O, Meerkerk GJ. Compulsive use of social networking sites in Belgium: prevalence, profile, and the role of attitude toward work and school. Cyberpsychol Behav Soc Netw. 2014;17:166–171. doi: 10.1089/cyber.2013.0029. [DOI] [PubMed] [Google Scholar]

26. Shensa A, Escobar-Viera CG, Sidani JE, Bowman ND, Marshal MP, Primack BA. Problematic social media use and depressive symptoms among U.S. young adults: A nationally-representative study. Soc Sci Med. 2017;182:150–157. doi: 10.1016/j.socscimed.2017.03.061. [DOI] [PMC free article] [PubMed] [Google Scholar]

27. Lin LY, Sidani JE, Shensa A, Radovic A, Miller E, Colditz JB, Hoffman BL, Giles LM, Primack BA. Association between social media use and depression among U.S. young adults. Depress Anxiety. 2016;33:323–331. doi: 10.1002/da.22466. [DOI] [PMC free article] [PubMed] [Google Scholar]

28. Balta S, Emirtekin E, Kircaburun K, Griffiths MD. Neuroticism, trait fear of missing out, and phubbing: the mediating role of state fear of missing out and problematic Instagram use. Int J Ment Health Addict. 2020;18:628–639. [Google Scholar]

29. Blackwell D, Leaman C, Tramposch R, Osborne C, Liss M. Extraversion, neuroticism, attachment style and fear of missing out as predictors of social media use and addiction. Pers Individ Dif. 2017;116:69–72. [Google Scholar]

30. Arain M, Haque M, Johal L, Mathur P, Nel W, Rais A, Sandhu R, Sharma S. Maturation of the adolescent brain. Neuropsychiatr Dis Treat. 2013;9:449–461. doi: 10.2147/NDT.S39776. [DOI] [PMC free article] [PubMed] [Google Scholar]

31. Simons JS, Gaher RM, Correia CJ, Hansen CL, Christopher MS. An affective-motivational model of marijuana and alcohol problems among college students. Psychol Addict Behav. 2005;19:326–334. doi: 10.1037/0893-164X.19.3.326. [DOI] [PubMed] [Google Scholar]

32. Martens MP, Neighbors C, Lewis MA, Lee CM, Oster-Aaland L, Larimer ME. The roles of negative affect and coping motives in the relationship between alcohol use and alcohol-related problems among college students. J Stud Alcohol Drugs. 2008;69:412–419. doi: 10.15288/jsad.2008.69.412. [DOI] [PMC free article] [PubMed] [Google Scholar]

33. Hogan B. Social media giveth, social media taketh away: Facebook, friendships, and APIs. Int J Commun. 2018;12:592–611. [Google Scholar]

34. Markham H. The Epidemic of Loneliness. Time Inc., 2018: 18-21. [Google Scholar]

35. Culnan MJ, Markus M. Information technologies. In: Jablin FM, Putnam LL, Roberts KH, Porter LW, editors. Handbook of organizational communication: An interdisciplinary perspective. Thousand Oaks: SAGE, 1987; 420–443. [Google Scholar]

36. Lou LL, Yan Z, Nickerson A, McMorris R. An Examination of the Reciprocal Relationship of Loneliness and Facebook Use among First-Year College Students. JECR. 2012;46:105–117. [Google Scholar]

37. Lange PG. Publicly Private and Privately Public: Social Networking on YouTube. J Comput Mediat Commun. 2007;13:361–380. [Google Scholar]

38. Jones MA, Mothersbaugh DL, Beatty SE. Switching barriers and repurchase intentions in services. J Retailing. 2000;76:259–274. [Google Scholar]

39. Adams SK, Murdock KK, Daly-Cano M, Rose M. Sleep in the Social World of College Students: Bridging Interpersonal Stress and Fear of Missing Out with Mental Health. Behav Sci (Basel) 2020;10 doi: 10.3390/bs10020054. [DOI] [PMC free article] [PubMed] [Google Scholar]

40. Li L, Griffiths MD, Mei S, Niu Z. Fear of Missing Out and Smartphone Addiction Mediates the Relationship Between Positive and Negative Affect and Sleep Quality Among Chinese University Students. Front Psychiatry. 2020;11:877. doi: 10.3389/fpsyt.2020.00877. [DOI] [PMC free article] [PubMed] [Google Scholar]

41. Shoval D, Tal N, Tzischinsky O. Relationship of smartphone use at night with sleep quality and psychological well-being among healthy students: A pilot study. Sleep Health. 2020;6:495–497. doi:

10.1016/j.sleh.2020.01.011. [DOI] [PubMed] [Google Scholar]

42. Scott H, Woods HC. Fear of missing out and sleep: Cognitive behavioural factors in adolescents' nighttime social media use. J Adolesc. 2018;68:61–65. doi: 10.1016/j.adolescence.2018.07.009. [DOI] [PubMed] [Google Scholar]

43. Mortazavi SAR, Parhoodeh S, Hosseini MA, Arabi H, Malakooti H, Nematollahi S, Mortazavi G, Darvish L, Mortazavi SMJ. Blocking Short-Wavelength Component of the Visible Light Emitted by Smartphones' Screens Improves Human Sleep Quality. J Biomed Phys Eng. 2018;8:375–380. [PMC free article] [PubMed] [Google Scholar]

44. Alt D. Students' Wellbeing, Fear of Missing out, and Social Media Engagement for Leisure in Higher Education Learning Environments. Curr Psychol. 2018;37:128–138. [Google Scholar]

45. Azizi SM, Soroush A, Khatony A. The relationship between social networking addiction and academic performance in Iranian students of medical sciences: a cross-sectional study. BMC Psychol. 2019;7:28. doi: 10.1186/s40359-019-0305-0. [DOI] [PMC free article] [PubMed] [Google Scholar]

46. Ophir E, Nass C, Wagner AD. Cognitive control in media multitaskers. Proc Natl Acad Sci U S A. 2009;106:15583–15587. doi: 10.1073/pnas.0903620106. [DOI] [PMC free article] [PubMed] [Google Scholar]

47. Hur JL, Gupta M. Growing up in the web of social networking: Adolescent development and social media. Adolesc Psychiatry. 2013;3:233–244. [Google Scholar]

48. Jha RK, Shah DK, Basnet S, Paudel KR, Sah P, Sah AK, Adhikari K. Facebook use and its effects on the life of health science students in a private medical college of Nepal. BMC Res Notes. 2016;9:378. doi: 10.1186/s13104-016-2186-0. [DOI] [PMC free article] [PubMed] [Google Scholar]

49. Vaidya CJ. Neurodevelopmental abnormalities in ADHD. Curr Top Behav Neurosci. 2012;9:49–66. doi: 10.1007/7854_2011_138. [DOI] [PMC free article] [PubMed] [Google Scholar]

50. Hoza B. Peer functioning in children with ADHD. J Pediatr Psychol. 2007;32:655–663. doi: 10.1093/jpepsy/jsm024. [DOI] [PubMed] [Google Scholar]

51. Humphreys KL, Galán CA, Tottenham N, Lee SS. Impaired Social Decision-Making Mediates the Association Between ADHD and Social Problems. J Abnorm Child Psychol. 2016;44:1023–1032. doi: 10.1007/s10802-015-0095-7. [DOI] [PMC free article] [PubMed] [Google Scholar]

52. Weinstein E. Adolescent differential responses to social media browsing: Exploring causes and consequences for intervention. Comput Human Behav. 2017;76:3960405. [Google Scholar]

53. Booker CL, Skew AJ, Kelly YJ, Sacker A. Media Use, Sports Participation, and Well-Being in Adolescence: Cross-Sectional Findings From the UK Household Longitudinal Study. Am J Public Health. 2015;105:173–179. doi: 10.2105/AJPH.2013.301783. [DOI] [PMC free article] [PubMed] [Google Scholar]

54. Loh K, Redd S. Understanding and preventing computer vision syndrome. Malays Fam Physician. 2008;3:128–130. [PMC free article] [PubMed] [Google Scholar]

55. Damodaran KK, Sharma V, Purushothaman S. Relationship between the hand discomfort with the dimensions of hand and touch screen mobiles. Drug Invention Today. 2019;12:537–540. [Google Scholar]

56. Santos VA, Freire R, Zugliani M, Cirillo P, Santos HH, Nardi AE, King AL. Treatment of Internet Addiction with Anxiety Disorders: Treatment Protocol and Preliminary Before-After Results Involving Pharmacotherapy and Modified Cognitive Behavioral Therapy. JMIR Res Protoc. 2016;5:e46. doi: 10.2196/resprot.5278. [DOI] [PMC free article] [PubMed] [Google Scholar]

57. Alutaybi A, Al-Thani D, McAlaney J, Ali R. Combating Fear of Missing Out (FoMO) on Social Media: The FoMO-R Method. Int J Environ Res Public Health. 2020;17 doi: 10.3390/ijerph17176128. [DOI] [PMC free article] [PubMed] [Google Scholar]

58. Vandoninck S, d'Haenens L, Roe K. Online risks: Coping strategies of less resilient children and teenagers across Europe. J Child Media. 2013;7:60–78. [Google Scholar]

59. Butler G, Gelder M, Hibbert G, Cullington A, Klimes I. Anxiety management: developing effective strategies. Behav Res Ther. 1987;25:517–522. doi: 10.1016/0005-7967(87)90059-3. [DOI] [PubMed] [Google Scholar]

60. Slutske WS. Natural recovery and treatment-seeking in pathological gambling: results of two U.S. national surveys. Am J Psychiatry. 2006;163:297–302. doi: 10.1176/appi.ajp.163.2.297. [DOI] [PubMed] [Google Scholar]

61. Centers for Disease Control and Prevention. Infographics - Screen Time vs. Lean Time. Available from: http://www.cdc.gov/nccdphp/dnpao/multimedia/infographics/getmoving.html .

62. Restrepo A, Scheininger T, Clucas J, Alexander L, Salum GA, Georgiades K, Paksarian D, Merikangas KR, Milham MP. Problematic internet use in children and adolescents: associations with psychiatric disorders and impairment. BMC Psychiatry. 2020;20:252. doi: 10.1186/s12888-020-02640-x. [DOI] [PMC free article] [PubMed] [Google Scholar]

63. Pulido CM, Redondo-Sama G, Sordé-Martí T, Flecha R. Social impact in social media: A new method to evaluate the social impact of research. PLoS One. 2018;13:e0203117. doi: 10.1371/journal.pone.0203117. [DOI] [PMC free article] [PubMed] [Google Scholar]

64. González-Bueso V, Santamaría JJ, Fernández D, Merino L, Montero E, Ribas J. Association between Internet Gaming Disorder or Pathological Video-Game Use and Comorbid Psychopathology: A Comprehensive Review. Int J Environ Res Public Health. 2018;15 doi: 10.3390/ijerph15040668. [DOI] [PMC free article] [PubMed] [Google Scholar]

65. Smith RH, Kim SH. Comprehending envy. Psychol Bull. 2007;133:46–64. doi: 10.1037/0033-2909.133.1.46. [DOI] [PubMed] [Google Scholar]

66. Lin R, Utz S. The emotional responses of browsing Facebook: Happiness, envy, and the role of tie strength. Comput Human Behav. 2015;52:29–38. doi: 10.1016/j.chb.2015.04.064. [DOI] [PMC free article] [PubMed] [Google Scholar]

67. van de Ven N, Zeelenberg M, Pieters R. Leveling up and down: the experiences of benign and malicious envy. Emotion. 2009;9:419–429. doi: 10.1037/a0015669. [DOI] [PubMed] [Google Scholar]

68. Tandoc EC, Ferrucci P, Duffy M. Facebook use, envy, and depression among college students: Is facebooking depressing? Comput Human Behav. 2015;43:139–146. [Google Scholar]

69. Appel H, Crusius J, Gerlach AL. Social comparison, envy, and depression on facebook: A study looking at the effects of high comparison standards on depressed individuals. J Soc Clin Psychol. 2015;34:277–289. [Google Scholar]

70. HealthyChildren Family Media Use Plan. Available from: http://www.healthychildren.org/English/media/Pages/default.aspx#wizard .

---

Articles from World Journal of Clinical Cases are provided here courtesy of **Baishideng Publishing Group Inc**